Then $\overrightarrow{AP}$ is perpendicular to $\vec{n}$.

So, $\overrightarrow{AP}\cdot\vec{n}=0$ which gives $(\vec{r}-\vec{a})\cdot\vec{n}=0$. ... (1)

which is the vector form of the equation of a plane passing through a point with position vector $\vec{a}$ and perpendicular to $\vec{n}$.

Note

$$(\vec{r}-\vec{a})\cdot\vec{n}=0 \Rightarrow \vec{r}\cdot\vec{n}=\vec{a}\cdot\vec{n} \Rightarrow \vec{r}\cdot\vec{n}=q, \text{ where } q=\vec{a}\cdot\vec{n}.$$

(b) Cartesian form of equation

If a,b,c are the direction ratios of $\vec{n}$, then we have $\vec{n}=a\hat{i}+b\hat{j}+c\hat{k}$. Suppose, A is (x_1,y_1,z_1) then equation (1) becomes $((x-x_1)\hat{i}+(y-y_1)\hat{j}+(z-z_1)\hat{k})\cdot(a\hat{i}+b\hat{j}+c\hat{k})=0$. That is,

$$a(x-x_1)+b(y-y_1)+c(z-z_1) = 0$$

which is the Cartesian equation of a plane, normal to a vector with direction ratios a,b,c and passing through a given point (x_1,y_1,z_1).

6.8.3 Intercept form of the equation of a plane

Let the plane $\vec{r}\cdot\vec{n}=q$ meets the coordinate axes at A,B,C respectively such that the intercepts on the axes are $OA=a$, $OB=b$, $OC=c$. Now position vector of the point A is $a\hat{i}$. Since A lies on the given plane, we have $a\hat{i}\cdot\vec{n}=q$ which gives $\hat{i}\cdot\vec{n}=\frac{q}{a}$. Similarly, since the vectors $b\hat{j}$ and $c\hat{k}$ lie on the given plane, we have $\hat{j}\cdot\vec{n}=\frac{q}{b}$ and $\hat{k}\cdot\vec{n}=\frac{q}{c}$. Substituting $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$ in $\vec{r}\cdot\vec{n}=q$, we get $x\hat{i}\cdot\vec{n}+y\hat{j}\cdot\vec{n}+z\hat{k}\cdot\vec{n}=q$. So $x\left(\frac{q}{a}\right)+y\left(\frac{q}{b}\right)+z\left(\frac{q}{c}\right)=q$.

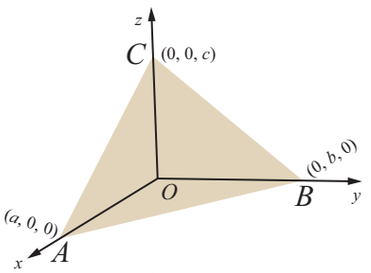


Fig. 6.26

Dividing by q, we get, $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$. This is called the **intercept form** of equation of the plane having intercepts a,b,c on the x,y,z axes respectively.

Theorem 6.16

The general equation $ax+by+cz+d=0$ of first degree in x,y,z represents a plane.

Proof

The equation $ax+by+cz+d=0$ can be written in the vector form as follows

$(x\hat{i}+y\hat{j}+z\hat{k})\cdot(a\hat{i}+b\hat{j}+c\hat{k})=-d$ or $\vec{r}\cdot\vec{n}=-d$.

Since this is the vector form of the equation of a plane in standard form, the given equation $ax+by+cz+d=0$ represents a plane. Here $\vec{n}=a\hat{i}+b\hat{j}+c\hat{k}$ is a vector normal to the plane. ■

Note

In the general equation $ax+by+cz+d=0$ of a plane, a,b,c are direction ratios of the normal to the plane.

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